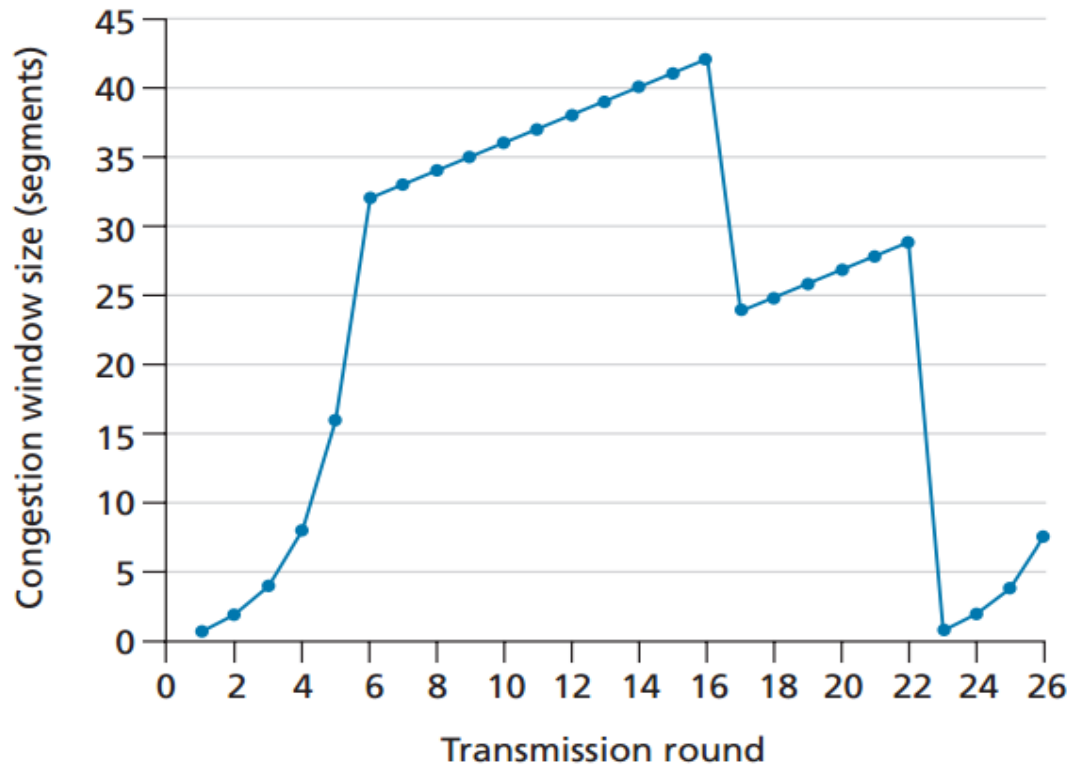


Class Tasks

TCP Reno is the protocol



- Identify the intervals of time when TCP slow start is operating.
- Identify the intervals of time when TCP congestion avoidance is operating.
- After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?
- What is the initial value of ssthresh at the first transmission round?
- What is the value of ssthresh at the 18th transmission round?
- What is the value of ssthresh at the 24th transmission round?
- h)** During what transmission round is the 70th segment sent?
- i) Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and of ssthresh?
- j) Suppose TCP Tahoe is used (instead of TCP Reno), and assume that triple duplicate ACKs are received at the 16th round. What are the ssthresh and the congestion window size at the 19th round?
- k) Again suppose TCP Tahoe is used, and there is a timeout event at 22nd round. How many packets have been sent out from 17th round till 22nd round, inclusive?

Answers:

- a. 1-6, 23 onwards
- b. 7-16, 17-22
- c. Duplicate Ack
- d. Timeout
- e. 32
- f. 24
- g. 14
- h. 1:1 (1) → 2:2 (3) → 3:4 (7) → 4:8 (15) → 5:16 (31) → 6:32 (63) → 7: 33 (96).
The count passes 70 in round 7, so the 70th segment is sent during round 7.
- i. $ssthresh=8/2=4$ $cwnd=4$
- j. $ssthresh=cwnd/2 = 21 \rightarrow cwnd=1 \rightarrow cwnd = 4$
- k. $ssthresh=cwnd/2=14 \rightarrow cwnd=1 \rightarrow 1+2+4+8+16+28=59$

Consider transferring an enormous file of L bytes from Host A to Host B. Assume an MSS of 536 bytes.

- a. What is the maximum value of L such that TCP sequence numbers are not exhausted? Recall that the TCP sequence number field has 4 bytes.
- b. For the L you obtain in (a), find how long it takes to transmit the file. Assume that a total of 66 bytes of transport, network, and data-link header are added to each segment before the resulting packet is sent out over a 155 Mbps link. Ignore flow control and congestion control.

Solution:

- a. $L_{max}=2^{(4*8)}-1=$ sequence numbers
- b. $N=\lceil L_{max}/MSS \rceil = 8,012,000$ segments
total per segment = 536+66=602 bytes
 $602 \times 8 = 4,816$ bits per packet
Total bits = $N \times 4,816 = 8.012 \times 10^6 \times 4,816 = 3.86 \times 10^{10}$ bits (≈ 38.6 Gb)
Time = $155 \times 10^6 \times 3.86 \times 10^{10} = 249$ seconds

Computer Networks

Quiz 4

Group A

Choose the most appropriate. Each MCQ carry 1 mark.

Total: 20 marks

1. TCP assigns sequence numbers to _____.
A. Segments B. Packets C. Bytes in the data stream D. ACKs
2. TCP's cumulative ACK acknowledges _____.
A. Only the last segment B. All bytes up to the first missing byte C. Each byte separately D. Next two expected bytes

3. The TCP field advertising available buffer space is _____.
A. cwnd B. rwnd C. seq # D. MSS
4. TCP keeps its connection state in _____.
A. Routers B. Switches C. Intermediate nodes D. End hosts
5. RTT estimation in TCP uses _____.
A. Fixed constant B. Last sample only C. Exponential weighted average D. Median of samples
6. If timeout interval is too short, TCP will _____.
A. Retransmit prematurely B. Delay transmission C. Drop ACKs D. Reset connection
7. Fast Retransmit is triggered by _____.
A. Timer expiry B. Three duplicate ACKs C. Checksum error D. RTT variance
8. During the 3-way handshake, TCP achieves _____.
A. Data transfer only B. MSS negotiation only C. Mutual sequence-number synchronization D. Router state setup
9. When receiver buffers fill up, TCP uses _____.
A. Multiplexing B. Congestion control C. Fragmentation D. Flow control
10. TCP is full-duplex because _____.
A. It sends and receives simultaneously on one connection
B. It uses two ports per direction
C. It doubles throughput
D. It splits stream into messages
11. Why does TCP choose random initial sequence numbers (ISN)?
A. To synchronize window sizes
B. To prevent old duplicate segments from previous connections being accepted
C. To ensure higher throughput
D. To match IP packet IDs
12. Host A received bytes 0–535 and 900–1000 but not 536–899. What ACK number is sent?
A. 1001 B. 900 C. 536 D. 535
13. Why does TCP average multiple SampleRTT values instead of using the latest one?
A. To reduce impact of random RTT fluctuations and smooth timeout estimates
B. To make timers update faster
C. Because old RTT values are more accurate
D. To match DevRTT exactly
14. How does TCP react differently to a timeout compared to receiving duplicate ACKs?
A. Timeout causes a delayed retransmission after the timer expires, while duplicate ACKs trigger an immediate fast retransmission.
B. Timeout closes the connection, while duplicate ACKs are ignored.
C. Both timeout and duplicate ACKs cause the connection to reset immediately.
D. Neither event leads to retransmission.
15. What does TCP's flow control mechanism prevent and how?
A. Prevents network congestion via cwnd
B. Prevents receiver buffer overflow via advertised rwnd
C. Prevents packet corruption via checksum
D. Prevents out-of-order delivery via ACK delay

Answers: C, B, B, D, C, A, B, C, D, A, B, C, A, A, B

Scenario

A TCP sender communicating between two hosts measures a sudden increase in network delay. Its EstimatedRTT is **currently** 100 ms, but the latest SampleRTT jumped to 120 ms due to temporary congestion. The sender's old DevRTT is 5 ms, and it updates this using $\beta = 0.25$. After updating DevRTT, the sender recalculates the Timeout Interval using the formula. What will be the new Timeout Interval (in ms) used by TCP after this update?

Solution:

$$\begin{aligned}\text{DevRTT} &= (1 - 0.25) \times 5 + 0.25 \times |120 - 100| \\ &= 3.75 + 5 = 8.75 \text{ ms}\end{aligned}$$

$$\text{TimeoutInterval} = 100 + 4 \times 8.75 = 135 \text{ ms}$$

✓ Final Answer: 135 ms

Computer Networks

Quiz 3

Group A

1. If the network layer already ensures delivery of packets between hosts, why do we still need the transport layer?
 - A. To speed up packet delivery between routers
 - B. To guarantee physical-layer transmission
 - C. To reduce network congestion at routers
 - D. To provide process-level communication and reliability features
2. When a UDP segment arrives at a destination host, demultiplexing is done using:
 - A. Source IP address only
 - B. Source and destination port numbers
 - C. Destination port number only
 - D. Source and destination IP addresses
3. Two different reliable data-transfer protocols handle errors differently:
 - Protocol A: uses ACKs/NAKs.
 - Protocol B: uses only ACKs but includes sequence numbers.If the channel flips bits in an ACK, which protocol avoids ambiguity better and why?
 - A. Protocol A — because it explicitly sends NAKs
 - B. Protocol B — because sequence numbers help detect duplicates and lost ACKs

- C. Both equally — corruption handling is identical
 - D. Neither — both fail if the ACK is lost
4. In Go-Back-N protocol, when packet n times out:
 - A. Packet n and all higher-numbered packets are retransmitted
 - B. Only packet n is retransmitted
 - C. All packets up to n are retransmitted
 - D. Sender waits for ACK for all packets before retransmission
 5. Which statement best describes Selective Repeat compared to Go-Back-N?
 - A. SR retransmits all unACKed packets at once
 - B. SR acknowledges and retransmits packets individually
 - C. SR does not buffer out-of-order packets
 - D. SR requires smaller sequence number space

Scenario: A company is using a Go-Back-N protocol for transmission. A sender wants to transmit 10 packets (50-59) to receiver with a window size of 4 (50-53). Due to network congestion, packets 53 and packet 56 are lost.

6. When packet 3 is lost, what will happen after the timeout occurs at the sender (assuming cumulative ACKs are used)?
 - A. The sender will retransmit only packet 53.
 - B. The sender will retransmit packets 53,56,57,58.
 - C. The sender will retransmit packets 53,54,55,56.
 - D. The sender will retransmit only packets that were not yet acknowledged individually.
7. After retransmitting due to the loss of packet 53, the sender successfully delivers packets 53, 54, 55, but then packet 56 is lost. What acknowledgment will the receiver keep sending?
 - A. ACK 56
 - B. ACK 55
 - C. ACK 52
 - D. ACK 53

Scenerio: A company upgrades its transmission system from Go-Back-N to Selective Repeat (SR) to improve efficiency. The sender wants to transmit 10 packets (50–59) using a window size of 4(50-53).

8. When packet 52 is lost, but packets 53 and 54 arrive correctly, what does the receiver do?
 - A. Discards packets 53 and 54 because they are out of order
 - B. Buffers packets 53 and 54 and keeps sending ACK 51
 - C. Buffers packets 53 and 54 and sends ACKs 53 and 54
 - D. Requests retransmission of all packets after packet 52

9. After packet 52 times out, the sender receives ACKs for packets 53 and 54. What does the sender do next?
- A. Retransmits only packet 52
 - B. Retransmits packets 52, 53, and 54
 - C. Waits for cumulative ACK up to 54
 - D. Retransmits packets 52, 55, and 56
10. After the receiver gets the retransmitted packet 52, what happens next?
- A. Receiver delivers packet 52 immediately but not 53 and 54
 - B. Receiver delivers packets 52, 53, 54 to application and sends ACK 52
 - C. Receiver delivers only up to packet 52
 - D. Receiver delivers packets 52, 53, 54 to application and sends ACK 54

Answers: D, C, B, A, B, C, B, C, A, B